SWAMI KESHVANAND INSTITUTE OF TECHNOLOGY, MANAGEMENT& GRAMOTHAN, JAIPUR

Subject: **Statistics and Probability Theory [Course Code: MAUL301CS/IT/DS/IOT/AI]**

Assignment: 2

SET-1

Q. No	**Question(s)**	BL	CO	MM
1	Prove that correlation coefficient is independent from change of origin and scale.	3	4	10
2	Calculate the Karl Pearson's coefficient of correlation of the following data <table><tr><td>x</td><td>25</td><td>27</td><td>30</td><td>35</td><td>33</td><td>28</td><td>36</td></tr><tr><td>y</td><td>19</td><td>22</td><td>27</td><td>28</td><td>30</td><td>23</td><td>28</td></tr></table>	3	4	10
3	Explain Birth- death Process	3	5	10
4	Explain queuing model $M\|M\|1 : \infty\|FIFO$	3	5	10

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Assignment:2
SET-2

Q. No	Question(s)	BL	CO	MM
1	Prove that the correlation coefficient lies between -1 and +1.	3	4	10
2	In a partially destroyed laboratory on record of an analysis of correlation data the following results only are legible, Var x = 9, regression equations: $8x - 10y + 66 = 0$, $40x - 18y = 214$ then find (i) the mean values of X and Y (ii) the coefficient of correlation between x and y (iii) the standard deviation of Y	3	4	10
3	Explain Discrete Parameter Birth Death Process	3	5	10
4	At a Petrol pump customers arrive in a Poisson process with average time of 7.5 minutes between two arrivals. The time interval between services at the pump is exponentially distributed with mean 3 minutes per customer. Determine (i) Average queue length and average number of customers in the system (ii) Time spent by a customer in the queue and in the system.	3	5	10

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Assignment:2

SET-3

Q. No	Question(s)	BL	CO	MM
1	For a bivariate distribution n = 18, $\sum x = 12$, $\sum y = 18$, $\sum xy = 48$, $\sum x^2 = 60$, $\sum y^2 = 96$. Find the equations of the lines of regression and r.	3	4	10
2	Find the normal equations of the best fitted second degree polynomial using method of least squares.	3	4	10
3	Explain pure birth process	3	5	10
4	Explain queuing model $M\|M\|1 : N\|FIFO$	3	5	10

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Assignment:2
SET-4

Q. No	Question(s)	BL	CO	MM
1	Show that in the case of perfect correlation two lines of regression coincide.	3	4	10
2	Two random variables have the least square regression lines with equations $3x + 2y - 26 = 0$ and $6x + y - 31 = 0$. Find the mean values and coefficient of correlation between x and y.	3	4	10
3	Explain Pure death process	3	5	10
4	Arrivals at a telephone booth are considered to be Poisson with an average time of 12 min between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean 4 min. i). Find the average number of persons waiting in the system ii). What is the probability that a person arriving at the booth will have to wait in the queue iii). What is the probability that it will take him more than 10 min altogether to wait for the phone and complete his call. iv). Estimate the fraction of the day when the phone will be in use. v). What is the average length of the queue that forms from time to time.	3	5	10

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Assignment:2

SET-5

<table>
<tr><th>Q. No</th><th>Question(s)</th><th>BL</th><th>CO</th><th>MM</th></tr>
<tr><td>1</td><td>Calculate the coefficient of correlation between x and y using following data:
<table>
<tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr>
<tr><td>Y</td><td>9</td><td>8</td><td>10</td><td>12</td><td>11</td><td>13</td><td>14</td><td>16</td><td>15</td></tr>
</table></td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>2</td><td>Fit a straight line to the following data:
<table>
<tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>6</td><td>8</td></tr>
<tr><td>Y</td><td>2.4</td><td>3</td><td>3.6</td><td>4</td><td>5</td><td>6</td></tr>
</table></td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>3</td><td>Explain queuing model $M|M|s : \infty|FIFO$</td><td>3</td><td>5</td><td>10</td></tr>
<tr><td>4</td><td>Traffic to a message switching center for one of the outgoing communication lines arrive in a random pattern at an average rate of 240 messages per minute. The line has a transmission rate of 800 characters per second. The message length distribution (including control characters) is approximately exponential with an average length of 176 characters. Calculate the following principal statistical measures of system performance, assuming that a very large number of message buffers are provided:
a. Average number of messages in the system
b. Average number of messages in the queue waiting to be transmitted.
c. Average time a message spends in the system.
d. Average time a message waits for transmission
e. Probability that 10 or more messages are waiting to be transmitted (not including the message being transmitted)</td><td>3</td><td>5</td><td>10</td></tr>
</table>

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Assignment:2

SET-6

Q. No	Question(s)	BL	CO	MM
1	Find r when it is given that the two regression coefficients are 0.8 and 1.2	3	4	10
2	Fit a parabola of the second degree to the following data: X: 1, 3, 5, 7, 9 Y: 2, 7, 10, 11, 9	3	4	10
3	Four counters are being run on the frontier of a country to check the passport and necessary papers of the tourists. The tourists choose counters at random. If the arrivals at the frontier is Poisson at the rate λ and the service time is exponential with parameter $\frac{\lambda}{2}$. What is the Steady State average at each Counter?	3	5	10
4	A call center with one operator has an incoming call rate of three calls per hour. On the average, the operator spends 15 minutes for each call. Answer the following questions: a. How busy is the operator? b. How likely is it that a customer will have to wait on hold? c. How long can we expect customers to have to wait on hold? d. What arrival rate can a single operator reasonably handle without causing excessive waiting (less than 5 minutes)?	3	5	10

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Assignment:2

SET-7

<table>
<tr><th>Q. No</th><th>Question(s)</th><th>BL</th><th>CO</th><th>MM</th></tr>
<tr><td>1</td><td>Define the coefficient of Correlation and Prove that for two independent variables r=0 but the converse is not true.</td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>2</td><td>Fit a parabola of the second degree to the following data:
<table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table></td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>3</td><td>Explain the queuing model $M|G|1$</td><td>3</td><td>5</td><td>10</td></tr>
<tr><td>4</td><td>A branch office of a large engineering firm has one on-line terminal that is connected to a central computer system during the normal eight-hour working day. Engineers, who work throughout the city, drive to the branch office to use the terminal to make routine calculations. Statistics collected over a period of time indicate that the arrival pattern of people at the branch office to use the terminal has a Poisson (random) distribution, with a mean of 10 people coming to use the terminal each day. The distribution of time spent by an engineer at a terminal is exponential, with a mean of 30 minutes. The branch office receives complains from the staff about the terminal service. It is reported that individuals often wait over an hour to use the terminal and it rarely takes less than an hour and a half in the office to complete a few calculations. The manager is puzzled because the statistics show that the terminal is in use only 5 hours out of 8, on the average. This level of utilization would not seem to justify the acquisition of another terminal. What insight can queuing theory provide?</td><td>3</td><td>5</td><td>10</td></tr>
</table>

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Assignment:2

SET-8

Q. No	Question(s)	BL	CO	MM
1	The joint Probability Distribution of X and Y is given below: <table><tr><td>Y/X</td><td>-1</td><td>1</td></tr><tr><td>0</td><td>1/8</td><td>3/8</td></tr><tr><td>1</td><td>2/8</td><td>2/8</td></tr></table>Find the correlation coefficient between X and Y.	3	4	10
2	Obtain the rank Correlation Coefficient for the following data: <table><tr><td>X</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>Y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>74</td></tr></table>	3	4	10
3	State and prove Chapman-Kolmogorov Theorem	3	5	10
4	Assume that you have a printer that can print an average file in two minutes. Every two and a half minutes a user sends another file to the printer. a. How long does it take before a user can get their output? b. To speed things up you can buy another printer that is exactly the same as the one you have. How long will it take for a user to get their files printed if you had two identical printers? c. Another solution is to replace the existing printer with one that can print a file in an average of one minute. How long does it take for a user to get their output with the faster printer?	3	5	10

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Assignment:2

SET-9

<table>
<tr><th>Q. No</th><th>Question(s)</th><th>BL</th><th>CO</th><th>MM</th></tr>
<tr><td>1</td><td>Obtain the Correlation Coefficient for the following data:
<table><tr><td>Y/X</td><td>0-8</td><td>8-16</td><td>16-24</td></tr><tr><td>1-5</td><td>2</td><td>0</td><td>4</td></tr><tr><td>5-9</td><td>3</td><td>2</td><td>2</td></tr><tr><td>9-13</td><td>2</td><td>5</td><td>1</td></tr></table></td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>2</td><td>Obtain the coefficient of Correlation for X, Y from the following data:
<table><tr><td>x</td><td>45</td><td>55</td><td>56</td><td>58</td><td>60</td><td>65</td><td>68</td><td>70</td><td>75</td><td>80</td><td>85</td></tr><tr><td>y</td><td>56</td><td>50</td><td>48</td><td>60</td><td>62</td><td>64</td><td>65</td><td>70</td><td>74</td><td>82</td><td>90</td></tr></table>
Obtain the equations of the lines of Regression.</td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>3</td><td>The tpm of two market products A and B is $\begin{bmatrix} .9 & .1 \\ .5 & .5 \end{bmatrix}$. These two brand have equal popularity. Find their steady state probabilities</td><td>3</td><td>5</td><td>10</td></tr>
<tr><td>4</td><td>Let there be an automobile inspection situation with three inspection stalls. Assume that cars wait in such a way that stall becomes vacant, the car at the head of the line pulls up to it .The station can accommodate almost four cars waiting at one time. The arrival pattern is Poisson with a mean of one car every minute during the peak hours. The service time is exponential with a mean of 6 minutes. Find the average number of customers in the system during peak hours the average waiting time and the average number per hour that cannot enter the station because of full capacity.</td><td>3</td><td>5</td><td>10</td></tr>
</table>

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Assignment:2

SET-10

<table>
<tr><th>Q. No</th><th>Question(s)</th><th>BL</th><th>CO</th><th>MM</th></tr>
<tr><td>1</td><td>Fit an exponential curve to the following data:
X: 1, 2, 3, 4, 5, 6, 7, 8
Y: 1, 1.2, 1.8, 2.5, 3.6, 4.7, 6.6, 9.1</td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>2</td><td>Obtain the rank Correlation Coefficient for the following data:
X: 85, 74, 85, 50, 65, 78, 74, 60, 74, 90
Y: 78, 91, 78, 58, 60, 72, 80, 55, 68, 70</td><td>3</td><td>4</td><td>10</td></tr>
<tr><td>3</td><td>Explain queuing model $M|M|s : N|FIFO$</td><td>3</td><td>5</td><td>10</td></tr>
<tr><td>4</td><td>A Supermarket has two girls attending to sales at the counters. If the service time for each customer is exponential with mean 4 minutes and if the people arrive in Poisson fashion at the rate of 10 per hour,
a) What is the Probability that a customer has to wait for service?
b) What is the expected percentage of idle time for each girl?
c) If the customer has to wait in the queue , what is the expected length of his waiting time?</td><td>3</td><td>5</td><td>10</td></tr>
</table>